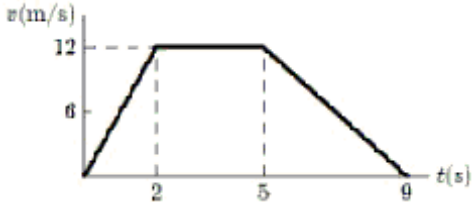
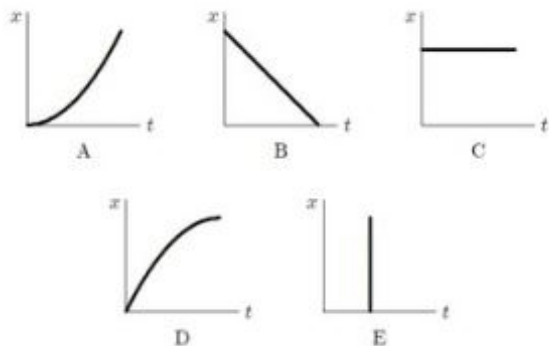


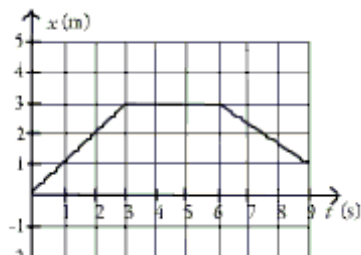
- 1) A car moving with an initial velocity of 25 m/s north has a constant acceleration of 3 m/s² south. After 6 seconds, its velocity will be: (hint: south is negative) 1) _____
 A) 43m/s North B) 7 m/s south C) 20 m/s North D) 7m/s North
- 2) If a stone falls to the bottom of a mineshaft in 6 seconds, then the depth of the shaft is about 2) _____
 A) 180 m. B) more than 200 m.
 C) 120 m. D) 60 m.
- 3) Which of those quantities is a scalar? 3) _____
 A) acceleration B) displacement C) distance D) velocity
- 4) A gun with a muzzle velocity of 100 m/s is fired horizontally from a tower. Neglecting air resistance, how far downrange will the bullet be 1 second later? 4) _____
 (that is : find ΔX along horizontal)
 A) 490 m
 B) 50 m
 C) 98 m
 D) 100 m
 E) none of the above
- 5) A kid goes around a circular path (circle) at a speed of 1m/s . 5) _____
 What is the displacement covered in 1 hour
 A) 1m B) 0 C) 2600m D) 60m
- 6) Here is the graph of the velocity of a car as a function of time, Between 5 and 9 s, the car is 6) _____
- 
- $v(\text{m/s})$ is on the y-axis
- A) speeding up gaining 6 m/s every second B) speeding up at a rate of 3m/s/s
 C) 0 D) slowing down losing 3m/s every second
- 7) A ball is thrown straight upward from ground level with a speed of 18 m/s. How much time passes before the ball strikes the ground if we disregard air resistance? 7) _____
 A) 1.8 s B) 1.1 s C) 3.7 s D) 0.6 s
- 8) One half second after starting from rest, a freely falling object will have a speed of about 8) _____
 A) 5 m/s.
 B) 20 m/s.
 C) 2 m/s.
 D) 10 m/s.
 E) none of the above

- 9) You drop a Physics textbook and a paper on the **Moon**. Which one will reach the ground first? 9) _____
- A) It depends on the mass of the textbook
 B) The textbook
 C) The paper
 D) They both reach the ground at the same time

- 10) Which of the following five coordinate versus time graphs represents the motion of an object moving with a constant nonzero speed? (the graphs are position x as a function of time) 10) _____



- A) A B) B C) C D) E E) D
- 11) The figure shows the position of an object as a function of time, with all numbers accurate to two significant figures. Between time $t = 0.0$ s and time $t = 3.0$ s what is the velocity of the car? 11) _____



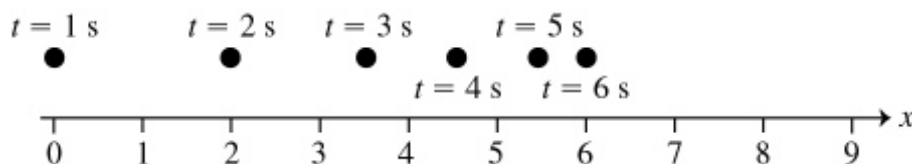
- A) 1 m/s B) 0 m/s C) 3 m/s D) 2 m/s
- 12) To determine the height of a bridge above the water, a person drops a stone and measures the time it takes for it to hit the water. If the time is 2.3 s, what is the height of the bridge? Neglect air resistance. 12) _____
- A) 10 m B) 52 m C) 14 m D) 26 m E) 32 m

- 13) The position x of a car moving along the x -axis is given by: $x = 3t + 5t^2$ at $t = 0$ the car is at $x = 0$. After 2s its velocity is ? 13) _____
- hint: The initial velocity is not 0.
 A) 0 B) 13m/s C) 23m/s D) 26m/s

- 14) A car, initially at rest, travels 20m in 4s along a straight line with constant acceleration. The acceleration of the car is: 14) _____
- A) 4.9 E) 9.8 B) 2.5 C) 1.3 D) 0.4

- 15) A car accelerates at 2 m/s². Assuming the car starts from rest, how much time does it need to accelerate to a velocity of 20 m/s? 15) _____
- A) 2 B) 20 C) 40 D) 10

- 16) A mosquito flying at 3 m/s that encounters a breeze blowing at 3 m/s in the opposite direction has a speed of _____
 A) 4 m/s. B) 3 m/s. C) 0 m/s. D) 6 m/s.
- 17) A racquetball strikes a wall with a speed of 30 m/s and rebounds in the opposite direction with a speed of 30 m/s. The collision takes 10 ms. What is the average acceleration of the ball during the collision with the wall? in magnitude (hint: 1ms = 0.001s) _____
 A) 2000 m/s² B) 0 m/s² C) 3000 m/s² D) 3 m/s² E) 6000 m/s²
- 18) You throw your physics book up in the air. At the top of the trajectory, _____
 A) only the acceleration is 0
 B) both the acceleration and velocity are 0
 C) only the velocity is 0
- 19) If a freely falling object were equipped with a speedometer, its speed reading would increase each second by about _____
 A) 15 m/s.
 B) 10 m/s.
 C) 5 m/s.
 D) depends on its initial speed
 E) a variable amount.
- 20) A hammer and a feather are dropped on the Moon from a height of 5m. _____
 A) both hit at the same time
 B) The hammer hits the ground first
 C) The feather hits the ground first
- 21) You throw a rock horizontally off a cliff with a speed of 20 m/s. The cliff is 100m high. How long for the rock to hit the ground. _____
 A) 10.5s B) 4.5s C) 20s D) 5s
- 22) The figure shows the position of an object (moving along a straight line) as a function of time. Which of the following statements about this object is true over the interval shown? _____



- A) The object is accelerating to the right.
 B) The average speed of the object is 1.0 m/s.
 C) The object is accelerating to the left.
 D) The acceleration of the object is in the same direction as its velocity.

23) Here are some equations and conversions.

$$v = v_0 + at$$

$$\Delta x = \left(\frac{v + v_0}{2}\right)t$$

$$\Delta x = v_0 t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a\Delta x$$

1 milli = 10^{-3} 1 kilo = 10^3 1 centi = 10^{-2} g = 9.8m/s/s 1hour = 3600s

Answer Key

Testname: EXAM1_22

- 1) D
- 2) A
- 3) C
- 4) D
- 5) B
- 6) D
- 7) C
- 8) A
- 9) D
- 10) B
- 11) A
- 12) D
- 13) C
- 14) B
- 15) D
- 16) C
- 17) E
- 18) C
- 19) B
- 20) A
- 21) B
- 22) C
- 23)